Phase 2 — Summary Quality Benchmark (Four-Dimension Framework)

Thesis anchor: §5.5 Future Works — "Evaluate AI summaries for relevance, coherence, factual consistency, and compression"; supports limitation #5 (summary consistency). **Date:** 2026-10-12 (FYP 2, Week 2) **Status:** complete (n = 10 paired articles, automatic metrics + cross-model LLM judging).

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1. Goal

Phase 1 (doc 01) measured only **length consistency**. This phase executes the full four-dimension evaluation the thesis promises, comparing the two models the system actually deploys:

Arm	Model	Hosting	Role
`llama31_v2`	llama3.1 (8B)	local Ollama (self-hosted)	primary; thesis "local AI" identity
`gptoss_v2`	gpt-oss:20b	Ollama Cloud (remote API)	former cloud fallback (benchmark-only arm; since replaced by `Beelzebub4883/llama3.2-3b-cloud` for testing)

The local-vs-cloud pairing doubles as evidence for the self-hosting future-work narrative: it measures what a self-hosted open SLM gives up (or keeps) versus a larger cloud-hosted open model behind the same prompt and cleanup pipeline.

Scope correction vs. earlier plan: MiMo (fallback_2) was dropped — it is not a model the bot is deployed with, and its API key is invalid (401). Benchmarking it would not be citable as production evidence.

2. Method

- **Fixture:** the same 50-article fixture as Phase 1 (`benchmark/articles_fixture.json`, seed 42); this phase uses the **first 10 articles** so the two arms are paired article-for-article. Inputs are RSS descriptions (see the Phase 1 proxy note).
- **Pipeline:** production prompt v2 + strict retry + `finalize_summary_candidate` cleanup via `scripts/benchmark_summaries.py` — identical to what the bot ships.
- **Quality metrics** (`scripts/benchmark_summary_quality.py`), per produced summary:
- *Compression* — summary_words / source_words.
- *Relevance proxy* — token-level ROUGE-L (precision / recall / F1) against the source.
- *Factual-consistency proxy* — fraction of summary content tokens, and of numeric tokens, present in the source (case-insensitive).
- *Coverage behavior* — abstention rate (`no_summary: true` rows).

LLM judging (1–5 rubric): cross-judge design so no model grades its own output — gpt-oss:20b (cloud) judges the 10 llama3.1 summaries; llama3.1 (local) judges the 5 gpt-oss summaries. Dimensions: relevance, coherence, factuality (rubric in script).

2.1 Operational finding: reasoning-model token budget (production-relevant)

`gpt-oss:20b` is a reasoning model: it spends output tokens on hidden reasoning before the visible answer. Under the production `OLLAMA_SUMMARY_NUM_PREDICT=384` cap, cloud responses returned **empty output** (`done_reason: length`, `eval_count: 384`) — i.e., the production cloud fallback as currently tuned cannot produce summaries at all. The benchmark arm therefore ran with a benchmark-only override of 1024 tokens (`scripts/_cloud_bench.py` sets the env var; production config and `.env` untouched). **Follow-up recommendation:** raise `OLLAMA_SUMMARY_NUM_PREDICT` (or make it model-dependent) before relying on the gpt-oss fallback in production.

Post-benchmark action (2026-11-08): `gpt-oss:20b` was replaced as the cloud testing fallback by `Beelzebub4883/llama3.2-3b-cloud`, because Qwen 2.5 was not available on Ollama Cloud and `gpt-oss` had moved to a paid tier. The results above remain valid as a one-time comparison of the local primary versus a larger cloud reasoning model.

3. Results

3.1 Automatic metrics (paired n = 10)

Metric	llama3.1 (local)	gpt-oss:20b (cloud)
Summaries produced	10 / 10	5 / 10
Abstention rate	0.0	0.50
Mean summary length (words)	20.4	19.0
Mean compression ratio (↓ = more compressed)	0.70	0.47
ROUGE-L F1 (vs source)	0.242	0.177
ROUGE-L recall	0.194	0.124
Content-token containment	0.476	0.443
Numeric-token containment	1.00	1.00

All 10 source texts are short (14–79 words, RSS descriptions), which caps how much compression is possible; the ratios should be read against that ceiling.

3.2 LLM-judged rubric (1–5, cross-judged)

Arm (judge)	n	Relevance	Coherence	Factuality
llama3.1 summaries, judged by gpt-oss:20b	10	3.9	4.7	3.3
gpt-oss:20b summaries, judged by llama3.1	5	5.0	5.0	5.0

Per-article detail is in `benchmark/judge_scores_20261012T085759Z.csv` (llama arm) and `benchmark/judge_scores_20261012T085616Z.csv` (gpt-oss arm).

4. Interpretation

1. **The headline finding is behavioral, not lexical.** The four factuality scores of 1 and one of 2 that gpt-oss gave llama3.1 all occur on the *thinnest inputs* (14–55-word descriptions: articles 423257, 427773, 431945, 412074; a 3 on the 28-word 406211). gpt-oss abstained on exactly those inputs. Conclusion: llama3.1 **always summarizes, and fabricates when the input is thin**; gpt-oss **declines to answer** instead. For a news-delivery bot, gpt-oss's precision-first behavior is the safer failure mode — but a 50% abstention rate on description-length inputs would mean half of summarized items deliver nothing, so neither behavior is directly acceptable.
2. **On adequate inputs both models are strong.** On the two articles where both models produced summaries (426583, 407993), the stricter judge scored llama3.1 5/5/5 and 5/4/5 (relevance/coherence/factuality). llama3.1's coherence stayed high (4.7) even on weak articles — the failure mode is factual, not linguistic.
3. **Judge-leniency asymmetry is real** (llama-as-judge gave all 5s, n = 5). Absolute scores should be treated cautiously; the reliable signal is the *strict judge's* relative pattern (factuality tracking input thinness) corroborated by the abstention data and the containment proxy.
4. **Numbers are safe in both models** (numeric containment 1.00 both arms) — dates, counts, and quantities in summaries were always traceable to the source.

4.1 Tie-back to thesis

• Supports **limitation #5** (summary consistency): the v2 prompt + retry (Phase 1) fixed length; this phase shows the remaining quality risk is *factual reliability on thin inputs*, not fluency.

• Feeds **future works**: stronger small models with built-in abstention calibration (Qwen / Ministral line, as §5.5 names) would let the bot keep local hosting while gaining gpt-oss-like refusal behavior; until then, the cloud fallback path needs the token-budget fix in §2.1.

5. Threats to validity

- **n = 10** paired articles; results are indicative, not statistical. The fixture has 50 articles if a larger run is wanted.
- **Input proxy:** RSS descriptions, not scraped full bodies; thin inputs are over-represented vs production. Abstention rates should be re-measured on full bodies.
- **Automatic proxies** reward extractive overlap; paraphrase and inference are under-credited (visible in llama's ROUGE-L being higher than its judged factuality).
- **Single judge per direction**, asymmetric judge leniency; no human gold standard yet (the planned ~150-respondent evaluation can incorporate one).
- **Benchmark-only token cap** for gpt-oss (§2.1): length results for the cloud arm reflect a 1024-token budget, not production's 384.

6. Reproduction

```
# 1. local arm (llama3.1, production settings) — already existed from Phase 1
python scripts/benchmark_summaries.py run --models llama3.1:latest `
--label llama31_v2 --offset 0 --limit 10 --out docs/fyp2/benchmark/llama31_v2.csv

# 2. cloud arm (gpt-oss:20b via configured Ollama Cloud fallback, 1024-token override)
python scripts/_cloud_bench.py run -- run --models gpt-oss:20b `
--label gptoss_v2 --prompt-version v2 --offset 0 --limit 10 --out docs/fyp2/benchmark/gptoss_v2.csv

# 3. quality metrics + cross-model judging (local Ollama must be up for the llama judge)
python scripts/benchmark_summary_quality.py `
docs/fyp2/benchmark/llama31_v2.csv docs/fyp2/benchmark/gptoss_v2.csv --judge
```

Artifacts (all under `docs/fyp2/benchmark/`): `gptoss_v2.csv`, `summary_quality_scores.csv`, `quality_metrics_20261012T085412Z.json`, `judge_scores_20261012T085616Z.csv`, `judge_scores_20261012T085759Z.csv`, `judge_metrics_20261012T090039Z.json`.